
Learning Conway’s Game of Life with CNN-Transformer Hybrids: Architecture Scaling, Exposure Bias, and Activation Inductive Bias

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Abstract

1 We study how a CNN-Transformer hybrid learns the one-step transition rule of
2 Conway’s Game of Life (GoL) on a 40×40 toroidal grid. Starting from a model
3 trained with pure teacher forcing, we show that autoregressive rollout suffers
4 from exposure bias, that a straight-through-estimator (STE) multi-step scheme
5 fails due to a systematic gradient bias, and that scheduled sampling resolves this
6 cleanly. We further show that once a model saturates its capacity, scaling the
7 training set five-fold *regresses* tail-case accuracy in a way invisible to aggregate
8 validation metrics, while scaling the model five-fold instead eliminates essentially
9 all remaining errors. Motivated by recent work on activation-function inductive
10 bias for cellular automata, we independently reproduce the finding that a learnable
11 second-degree polynomial activation lets a 25–34 parameter CNN learn GoL
12 reliably where ReLU fails, and report a negative result transferring this activation to
13 our larger multi-step model. Finally, we show that sinusoidal positional encodings
14 yield empirically more translation- equivariant patch embeddings than learned
15 ones.

16 1 Problem and architecture

17 GoL is a deterministic cellular automaton: each cell’s next state depends only on its own state and its
18 8-neighbor live count (B3/S23). We treat one GoL step as image-to-image prediction on a periodic
19 40×40 grid. Our core architecture is a CNN-Transformer hybrid: two circular-padded convolutions
20 extract local, translation-equivariant features respecting the torus topology; these are cut into non-
21 overlapping 4×4 patches, linearly projected, combined with a *learned* absolute positional embedding,
22 passed through a Transformer encoder for global mixing, and projected back to per-cell logits. GoL
23 is a natural stress test for learned representations: prior work shows plain feed-forward networks
24 struggle to learn it reliably at all [Springer and Kenyon, 2021], and the Transformer stage [Vaswani
25 et al., 2017] trades the CNN’s clean equivariance guarantee for long-range context (Section 3).

26 2 Training evolution: two failure modes and their fixes

27 **Exposure bias.** A model trained with pure single-step teacher forcing (V5) scores $> 99\%$ preci-
28 sion/recall on held-out single steps but collapses under autoregressive rollout, since it never sees
29 its own (imperfect) outputs during training. A straight-through-estimator (STE) multi-step scheme
30 intended to fix this instead **diverges** after epoch 5, which we trace to a systematic gradient bias from
31 the STE’s hard threshold. Replacing STE with *scheduled sampling* [Bengio et al., 2015] — gradually

Ver.	Key change	Val. loss	Prec./Rec.	Status
V5	Single-step teacher forcing	0.201	99.4% / 99.7%	Collapses under rollout
V6	+STE multi-step	0.234 ^a	92.7% / —	Diverges
V7	STE → scheduled sampling	0.201	99.7% / 100%	Stable
V8	+3× high-density upsample	0.201	100% / 99.9%	Best per-grid (291K params)
V9	+5× all data	0.201	99.7% / 99.6%	Regressed tail errors
V10	5.15× model size	0.199	100% / 100%	Zero errors , 2,500/2,500

Table 1: version summary. ^aV6’s best epoch before divergence.

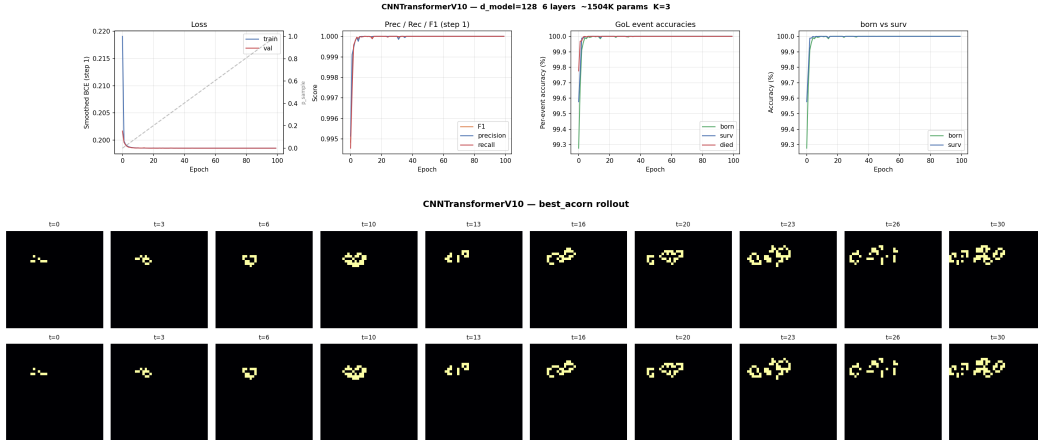


Figure 1: Top: V10 training curve. Bottom: a 30-step autoregressive rollout of the *acorn* methuselah — a 7-cell seed that grows into a large, long-lived chaotic pattern — true trajectory (top row) vs. V10’s autoregressive prediction (bottom row); V10 tracks the true trajectory exactly at every shown step, and makes zero errors across 2,500 held-out grids overall.

32 increasing the probability of feeding the model its own previous prediction — fixes this cleanly and
33 stably (V7).

34 **A hidden data-scaling regression.** Upsampling high-density configurations 3× (V8) improves
35 per-grid accuracy further (291K params). Naively scaling the *training set* 5× (V9) leaves aggregate
36 validation metrics statistically unchanged, but a density- stratified failure search (500 grids at each
37 of five densities, fixed seed across models) reveals V9’s error count at density 0.80 is nearly **seven**
38 **times worse** than V8’s (22,190 vs. 3,358 summed false positives/negatives) — invisible in aggregate
39 metrics alone. Instead of more data, we scale the *model*: V10 (5.15× V8’s size, 1.5M params, trained
40 from scratch) achieves **zero errors** across all 2,500 test grids at every density (Table 1, Figure 1). The
41 lesson: once capacity saturates, more data can silently trade average-case for tail-case performance,
42 detectable only by per-subgroup error auditing, not validation loss.

43 3 Embedding geometry: sinusoidal vs. learned positional encoding

44 A CNN feature map is a *physical* embedding: convolution’s translation-equivariance means shifting
45 the input shifts the feature map identically. A Transformer’s patch embeddings, after a learned
46 absolute positional embedding, have no such guarantee — each position slot is an independently
47 learned vector unrelated to its neighbors. We tested this empirically on two trained models differing
48 in positional encoding: one using a fixed 2D sinusoidal encoding (V2, whose angle-addition identities
49 give closed-form near-equivariant structure) and one using a learned absolute positional embedding
50 (V8). Measuring cosine similarity between patch embeddings of a grid and of translated copies of it,
51 V2’s post- Transformer embeddings stay far more similar under translation (mean cosine similarity
52 0.997) than V8’s (0.890), consistent with the theoretical prediction (Figure 2, left). We note an
53 acknowledged confound: V2 uses zero-padding versus V8’s circular padding, so this is not a fully
54 isolated ablation of positional encoding alone.

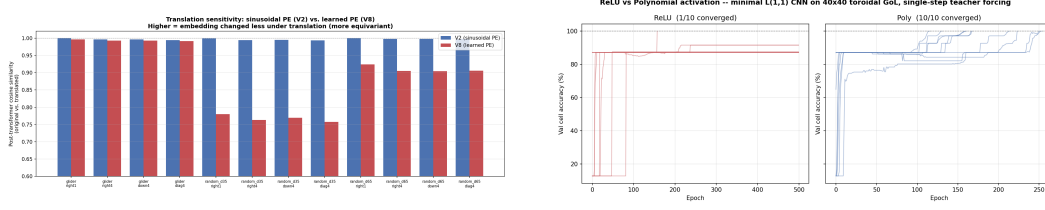


Figure 2: Left: post-Transformer embedding cosine similarity under translation, sinusoidal PE (V2) vs. learned PE (V8); higher means less change under translation, i.e. more equivariant. Right: validation accuracy over training, 10 seeds per activation, on a minimal 25/34- parameter CNN; polynomial activation converges reliably where ReLU does not.

4 Independent verification: polynomial activations for GoL

Motivated by a recent claim that a learnable second-degree polynomial activation $\phi(x) = w_0 + w_1x + w_2x^2$ lets a minimal CNN learn GoL reliably where ReLU fails almost completely [Ahmed and Davis, 2026], we independently reproduced this rather than taking it on faith. We built two CNNs matching the paper’s minimal architecture (two 1×1 convolutions plus one 3×3 circular-padded convolution), **identical except for the activation function**, matching the paper’s reported parameter counts exactly (ReLU: 25 params, polynomial: 34 params). Trained single-step, teacher-forcing, over 10 random seeds: ReLU converges to perfect validation accuracy in only 1/10 seeds (mean final accuracy 89.4%), while the polynomial activation converges in 10/10 seeds (100.0% mean accuracy), confirming the claim on our own toroidal GoL setup (Figure 2, right). We then attempted to transfer this activation into our larger multi-step model (same size as V8) by replacing GELU with polynomial activations in the CNN stage; this **failed**, diverging sharply after epoch 5 under scheduled sampling, likely because the unbounded w_2x^2 term compounds through the autoregressive feedback loop in a way it cannot in a single-step setting. Candidate fixes (a bounded/saturating variant, a separate learning rate for polynomial coefficients) were identified but not yet pursued.

5 Conclusion

Three practical lessons emerge. First, autoregressive exposure bias is best fixed with scheduled sampling rather than a hard differentiable relaxation (STE), which can introduce its own destabilizing gradient bias. Second, aggregate validation metrics can hide severe, density- specific regressions; per-subgroup error auditing is necessary once a model is scaled with more data rather than more capacity. Third, an activation function’s inductive-bias advantage demonstrated at minimal scale and single-step training does not automatically transfer to a larger, multi-step, autoregressively-trained model — the interaction between an activation’s stability properties and the training regime matters as much as the activation itself.

References

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